

Project Design Phase

Proposed Solution

Parameter	Details
Date	14 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	2 Marks

Proposed Solution Details

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Fruit vendors, farmers, and distributors often face difficulty in identifying rotten fruits among fresh ones. Manual inspection is time-consuming and sometimes inaccurate, leading to food wastage, financial losses, and reduced customer satisfaction.
2	Idea / Solution Description	Develop a Machine Learning-based Smart Sorting System that analyzes fruit images and classifies them as Fresh or Rotten using deep learning techniques. The system provides fast and accurate predictions through a simple web-based interface.
3	Novelty / Uniqueness	Uses image processing and deep learning models to automatically detect rotten fruits. The system reduces human error and enables faster sorting compared to traditional manual inspection methods.
4	Social Impact / Customer Satisfaction	Helps reduce food waste, improves fruit quality in markets, increases customer satisfaction, and supports farmers and vendors in maintaining better product standards.
5	Business Model (Revenue Model)	The solution can be deployed as a smart quality-check system for fruit markets, warehouses, and agricultural supply chains. It can be offered as a subscription-based software service or integrated into automated sorting machines.
6	Scalability of the Solution	The system can be expanded to detect multiple fruit types and diseases. With cloud deployment, it can support large-scale agricultural monitoring and automated sorting systems.